KAIZEN MATHS

Lesson starter

Hypothesis Testing and Inference

Binomial Tests

Name

S1

A sample of 21 gives 13 defective calculators. The original claim is that the defect rate is 0.4. Test at the 10% significance level whether the proportion has increased.

S2

In 20 trials, 13 customers selecting express delivery are observed. Test at the 1% significance level whether the proportion is different from 0.4.

S3

For a test of H_0: $p = 0.5$ against H_1: $p < 0.5$, use $X \sim B(27, 0.5)$ and a 5% significance level. Find the critical region and decide what happens if $x = 8$.

S4

A sample of 25 gives 15 students choosing the new revision platform. The original claim is that the long-term proportion is 0.4. Test at the 5% significance level whether the proportion has increased.

Show your working

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S4

A sample of 25 gives 15 students choosing the new revision platform. The original claim is that the long-term proportion is 0.4. Test at the 5% significance level whether the proportion has increased.

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KAIZEN MATHS

Exit ticket

Hypothesis Testing and Inference

Binomial Tests

Name _____

EA1

In 36 trials, 25 defective calculators are observed. Test at the 1% significance level whether the proportion is different from 0.6.

EA2

For a test of H_0: $p = 0.4$ against H_1: $p < 0.4$, use $X \sim B(26, 0.4)$ and a 10% significance level. Find the critical region and decide what happens if $x = 6$.

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Exit ticket

Hypothesis Testing and Inference

Binomial Tests

Name _____

EB1

A sample of 26 gives 4 customers selecting express delivery. The original claim is that the long-term proportion is 0.25. Test at the 10% significance level whether the proportion has increased.

KAIZEN MATHS

Exit ticket

Hypothesis Testing and Inference

Binomial Tests

Name _____

EA1

In 36 trials, 25 defective calculators are observed. Test at the 1% significance level whether the proportion is different from 0.6.

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For a test of H_0: $p = 0.4$ against H_1: $p < 0.4$, use $X \sim B(26, 0.4)$ and a 10% significance level. Find the critical region and decide what happens if $x = 6$.

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KAIZEN MATHS

Exit ticket

Hypothesis Testing and Inference

Binomial Tests

Name _____

EB1

A sample of 26 gives 4 customers selecting express delivery. The original claim is that the long-term proportion is 0.25. Test at the 10% significance level whether the proportion has increased.

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Teacher answers & working

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S1

A sample of 21 gives 13 defective calculators. The original claim is that the defect rate is 0.4. Test at the 10% significance level whether the proportion has increased.

Answer:

reject H_0

Working:

Let p be the proportion of calculators that are defective.

$H_0: p = 0.4$

$H_1: p > 0.4$

Under H_0, X ~ B(21, 0.4). The observed value is x = 13.

The alternative hypothesis tells us to use the upper tail.

p-value = P(X ≥ 13) = 0.0352

Compare the p-value with α = 0.1.

0.0352 ≤ 0.1, so reject H_0.

There is sufficient evidence at the 10% level to conclude that the proportion of calculators that are defective has increased.

S2

In 20 trials, 13 customers selecting express delivery are observed. Test at the 1% significance level whether the proportion is different from 0.4.

Answer:

do not reject H_0

Working:

Let p be the proportion of customers selecting express delivery.

$H_0: p = 0.4$

$H_1: p \neq 0.4$

Under H_0, X ~ B(20, 0.4). The observed value is x = 13.

Because H_1 uses ≠, this is a two-tailed test. We measure how extreme the result is in the observed tail and double that tail probability.

$P(X \leq 13) = 0.9935, P(X \geq 13) = 0.0210$

p-value = 2 × min(0.9935, 0.0210) = 0.0421

Compare the p-value with α = 0.01.

0.0421 > 0.01, so do not reject H_0.

There is insufficient evidence at the 1% level to conclude that the proportion of customers selecting express delivery is different from 0.4.

Teacher answers & working

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S3

For a test of H_0: p = 0.5 against H_1: p < 0.5, use X ~ B(27, 0.5) and a 5% significance level. Find the critical region and decide what happens if x = 8.

Answer:

reject H_0

Working:

H_0 fixes the probability at 0.5; H_1 tells us to use the lower tail.

$H_0: p = 0.5$

$H_1: p < 0.5$

Choose the smallest region whose probability is at most α = 0.05.

$P(X \leq 8) = 0.0261 \leq 0.05$

$P(X \leq 9) = 0.0610 > 0.05$

Critical region: X ≤ 8

The observed value is x = 8.

8 is in the critical region, so reject H_0.

The observed value lies in the critical region, so there is sufficient evidence that the proportion of calculators that are defective has decreased.

S4

A sample of 25 gives 15 students choosing the new revision platform. The original claim is that the long-term proportion is 0.4. Test at the 5% significance level whether the proportion has increased.

Answer:

reject H_0

Working:

Let p be the proportion of students who choose the new revision platform.

$H_0: p = 0.4$

$H_1: p > 0.4$

Under H_0, X ~ B(25, 0.4). The observed value is x = 15.

The alternative hypothesis tells us to use the upper tail.

p-value = P(X ≥ 15) = 0.0344

Compare the p-value with α = 0.05.

0.0344 ≤ 0.05, so reject H_0.

There is sufficient evidence at the 5% level to conclude that the proportion of students who choose the new revision platform has increased.

Teacher answers & working

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EA1

In 36 trials, 25 defective calculators are observed. Test at the 1% significance level whether the proportion is different from 0.6.

Answer:

do not reject H_0

Working:

Let p be the proportion of calculators that are defective.

$H_0: p = 0.6$

$H_1: p \neq 0.6$

Under H_0, X ~ B(36, 0.6). The observed value is x = 25.

Because H_1 uses ≠, this is a two-tailed test. We measure how extreme the result is in the observed tail and double that tail probability.

$P(X \leq 25) = 0.9096$, $P(X \geq 25) = 0.1620$

p-value = 2 × min(0.9096, 0.1620) = 0.3240

Compare the p-value with α = 0.01.

0.3240 > 0.01, so do not reject H_0.

There is insufficient evidence at the 1% level to conclude that the proportion of calculators that are defective is different from 0.6.

EA2

For a test of H_0: p = 0.4 against H_1: p < 0.4, use X ~ B(26, 0.4) and a 10% significance level. Find the critical region and decide what happens if x = 6.

Answer:

reject H_0

Working:

H_0 fixes the probability at 0.4; H_1 tells us to use the lower tail.

$H_0: p = 0.4$

$H_1: p < 0.4$

Choose the smallest region whose probability is at most α = 0.1.

$P(X \leq 6) = 0.0559 \leq 0.1$

$P(X \leq 7) = 0.1216 > 0.1$

Critical region: X ≤ 6

The observed value is x = 6.

6 is in the critical region, so reject H_0.

The observed value lies in the critical region, so there is sufficient evidence that the proportion of customers selecting express delivery has decreased.

Teacher answers & working

Hypothesis Testing and Inference · Binomial Tests

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EB1

A sample of 26 gives 4 customers selecting express delivery. The original claim is that the long-term proportion is 0.25. Test at the 10% significance level whether the proportion has increased.

Answer:

do not reject H_0

Working:

Let p be the proportion of customers selecting express delivery.

H_0: $p = 0.25$

H_1: $p > 0.25$

Under H_0, X ~ B(26, 0.25). The observed value is x = 4.

The alternative hypothesis tells us to use the upper tail.

p-value = P(X ≥ 4) = 0.9198

Compare the p-value with α = 0.1.

0.9198 > 0.1, so do not reject H_0.

There is insufficient evidence at the 10% level to conclude that the proportion of customers selecting express delivery has increased.